

enabled. In this article, I am only describing the ‘polling’ scheme.

Using @BotFather to create your bot

The steps in creating a Telegram bot are as follows:

1. Telegram has a bot named @BotFather which can create a bot for you. So, as a first step, go to your Telegram app and search for @BotFather.

2. Then send it the message /start. The scheme works like this—if it is a message without a leading forward slash (/), then it will be interpreted as a text message. However, if it has a leading forward slash, then it will be interpreted as the function / method. A Telegram bot cannot respond on its own (to reduce spamming by bots). It must receive a text message (one without a forward slash) or a command (one with a leading forward slash) to respond to such messages.

3. Now create a new bot using the command /newbot and give it an appropriate name, which must end in the word bot. @BotFather will send you a message, which will include a token. The token should look something like what’s shown below:

```
1104497538:AAEbXXXi PegAxUXXXX_
TevmyPG01GnXXLY
```

(Some letters have been replaced with X to preserve privacy.) Fig. 3 shows the process on the Telegram app.

4. Note that your newly created bot is hosted on the Telegram server, and you can now access your bot on a Web browser by using a command of the format:

```
https://api.telegram.org/bot<your_bot_token>/getme
```

The above URL has three parts: (1) `https://api.telegram.org/bot`; (2) `<your_bot_token>`; and (3) `/getme`. The first two parts of the URL will allow your browser to access your bot (which resides on the Telegram server) and the last part will give the appropriate command to your bot. Here the command is `/getme` and so it will provide details of your bot in the form of a JSON object (similar to a Python dictionary). Fig. 4 shows this.

Now there are two ways to interact with the bot. The first is, of course, the method we used earlier, where we used a Web browser and a URL of

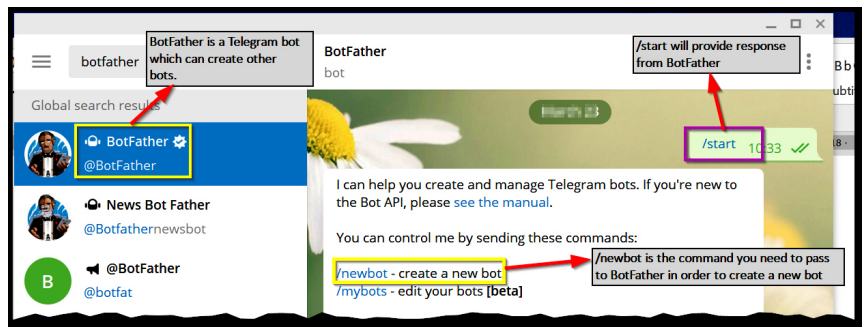


Fig. 2: Using the command ‘start’ on @BotFather

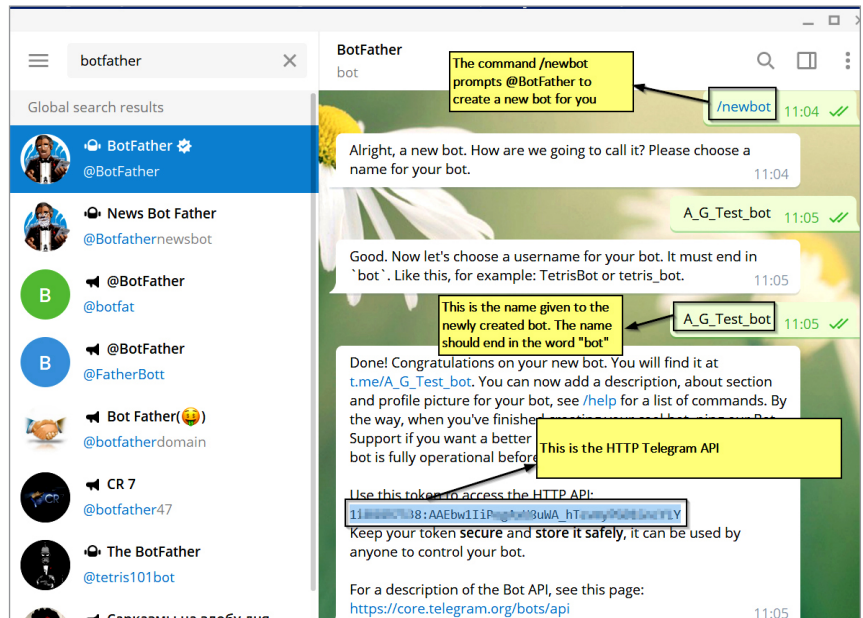


Fig. 3: Creating a new bot using @BotFather



Fig. 4: A Screenshot of the output on giving a command

the format `https://api.telegram.org/bot<your_bot_token>/getme`

There is, however, a second method in which we can use a Python script to access the messages/commands coming from the bot user to our bot hosted on the Telegram server. In the next section, we will look at creating a bot controller through a script written in Python.

Writing a Python script to create a bot controller using python-library-telegram

To create a bot controller script, we shall use a Python library `python-telegram-bot`; so download this using Pip:

```
pip install python-telegram-bot
```

This is shown in Fig. 5.

To write the Python script, you can use any popular IDE. Some common ones are PyCharm, Visual Studio Code, and Eclipse (with PyDev). You need to know some of the classes and methods of this library in order to truly understand how the code works. Remember, we are using the *polling* and not the *webhook* technique.

We shall use instances of (i.e., we will create objects of) two classes from the `telegram.ext` sub-module, namely, `telegram.ext.Updater` and `telegram.ext.Dispatcher`. The Updater class gets updates from Telegram and passes